

The β -oxidation of arachidonic acid and the synthesis of docosahexaenoic acid are selectively and consistently altered in skin fibroblasts from three Zellweger patients versus X-adrenoleukodystrophy, Alzheimer and control subjects

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Abstract

The β -oxidation of [³H] arachidonic acid (AA; 20:4 n-6) and the conversion of [1-¹⁴C]eicosapentaenoic acid (EPA, 20:5 n-3) to docosahexaenoic acid (DHA, 22:6 n-3) have been studied in skin fibroblasts from patients with inherited peroxisomal diseases, such as Zellweger (ZW) and X-linked adrenoleukodystrophy (X-ALD), from patients with Alzheimer's disease (AD), a non-inherited neuropathology, and from controls. EPA is not converted to DHA, while there is enhanced formation of the intermediate product 22:5 n-3 in ZW, when compared to X-ALD, AD and controls. We also confirmed that AA is not β -oxidized to 4,7,10-hexadecatrienoic acid (16:3), a metabolite produced by peroxisomes, while being more effectively converted to the elongation product 22:4, in ZW, in comparison to X-ALD, AD and controls. The data demonstrate a defect in DHA synthesis and in AA β -oxidation, and the occurrence of associated adaptative modifications in the metabolism of these long chain PUFA, in three Italian ZW patients. © 1998 Elsevier Science Ireland Ltd. All rights reserved

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Peroxisomal diseases are related to different degrees of peroxisomal deficiencies, involving one or more enzymes. These pathologies include the Zellweger disease (ZW), cerebro-hepato-renal syndrome, an autosomic recessive disorder with severe alteration in peroxisomal biogenesis [9], X-adrenoleukodystrophy (X-ALD), with a peroxisomal defect in the metabolism of very long chain fatty acids (VLCFA) [7,8].

Levels of docosahexaenoic acid (DHA, 22:6 n-3), are very low, especially in brain, in patients with severe peroxisomal disorders, such as X-ALD and the Zellweger syndrome [8,10], and some clinical symptoms are improved following the administration of this fatty acid [11]. More

recently, reduced synthesis of DHA from labelled precursors has been reported in ZW skin fibroblasts [12]. The 16 carbon trienoic acid (16:3) of the n-6 series has been identified as a new product of arachidonic acid (AA, 20:4 n-6) peroxisomal β -oxidation, released into the culture medium by various types of cells upon incubation with labeled AA, this metabolite is not produced by cells from ZW patients [5].

We have investigated AA β -oxidation [5] and DHA synthesis, both involving peroxisomes, in fibroblasts from three unrelated ZW and from one X-ALD patient, and compared the results with those obtained from two patients with Alzheimer's disease (AD), a non-inherited neurodegenerative pathology in which metabolic abnormalities have been demonstrated in peripheral tissues including fibroblasts [1] and from three normal subjects.

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[5,6,8,9,11,12,14,15-³H]Arachidonic acid (208 Ci/mmol) and [1-¹⁴C]arachidonic acid (53 mCi/mmol) were purchased from Amersham International (Buckinghamshire, UK). [1-¹⁴C]Eicosapentaenoic acid (EPA) (52 mCi/mmol) was from New England Nuclear (NEN; DuPont, Boston, MA, USA). The nucleosil C18 column (250 × 4.6 mm; particle size, 5 μm) used for the HPLC analysis of labeled fatty acid was from Alltech Associates (Deerfield, IL, USA).

The AD patients were diagnosed in the Alzheimer's Disease Unit of the IRCCS S. Giovanni di Dio, Brescia, Italy. The full clinical characterization of the AD patients from which the fibroblasts used in the present study were derived has been previously reported [6]. Skin fibroblasts from patients with the Zellweger syndrome and from one patient with X-ALD were kindly supplied by Dr. Bertagnolio (Istituto Nazionale Neurologico, Carlo Besta, Milan, Italy). The diagnoses of Zellweger and ALD were based on the analysis of VLCFA in plasma [8] and dihydroxyacetone phosphate acyltransferase (DHAPAT) in fibroblasts [14], and the data were compared with 20 controls of different ages (Table 1). The cells were grown in MEM containing 10% FCS, supplemented with non-essential amino acids, penicillin/streptomycin, tricine, and were maintained in a humidified atmosphere with 95% air and 5% CO₂ at 37°C.

Cultured cells at 80–100% confluence were incubated with either [³H]arachidonic acid (0.5 μCi/ml) up to 24 h or [1-¹⁴C]arachidonic acid (0.1 μCi/ml) for 18 h in MEM with 5% FCS in an incubator (37°C, 95% air and 5% CO₂).

At the end of the incubation the labeled medium was removed then cells were incubated for an additional period of 24 h in MEM containing 5% FCS. Aliquots of the lipid extracted from the medium were subjected to scintillation counting and an aliquot injected into the HPLC apparatus, connected with a radiodetector [5].

Docosatetraenoic acid (22:4 n-6) was identified by the use of a commercially available standard, whereas 22:5 n-6 was identified by using as reference this fatty acid isolated, after saponification and preparative HPLC, from lipids

Table 1

Values of 26:0, 26:0/22:0, 24:0/22:0 in plasma and dihydroxyacetone phosphate acyltransferase (DHAPAT) in fibroblasts of controls, ZW and ALD patients

Samples	Plasma			Fibroblast DHAPAT nmol/2 h per mg protein
	C26:0 mg/ml	C26:0/ C22:0 × 1000	C24:0/ C22:0 × 1000	
ZW 1	2.48	376	1976	0.12
ZW 2	4.50	611	2426	0.05
ZW 3	1.86	81	1188	0.18
ALD	1.06	53	1340	2.55
Controls (n = 20)	<0.60	<32	<1200	2.36 ± 0.7

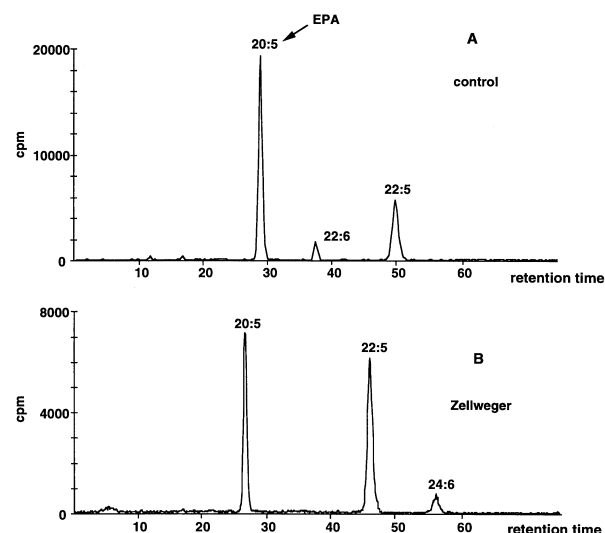


Fig. 1. (A,B) Typical HPLC profiles of products of eicosapentaenoic acid elongation and desaturation in lipids of normal skin fibroblasts (A) and fibroblasts from a Zellweger patient (B). Analyses were carried out by reverse phase HPLC of cell lipid fatty acid methyl esters after a 18 h incubation of fibroblasts with [1-¹⁴C]EPA (0.25 μCi/ml).

extracted from rat testis, an organ typically enriched in 22:5 n-6 [3].

Cells were incubated with [1-¹⁴C]EPA (0.25 μCi/ml) in MEM without FCS (18 h, 37°C). At the end of the incubation, cells were washed once with PBS, scraped from the dishes, lipids were extracted according to the method of Folch et al. [4] and fatty acid methyl esters were prepared by acidic transesterification and injected into the HPLC

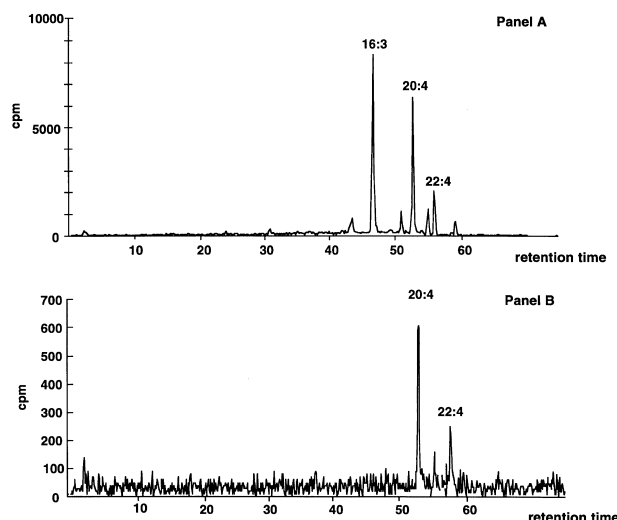


Fig. 2. (A,B) Typical HPLC profiles of radioactive products present in the medium after incubation of normal human skin fibroblasts with [³H]arachidonic acid (0.5 μCi/ml) (A) or [1-¹⁴C]arachidonic acid (0.1 μCi/ml) (B) for 18 h. After the labeled medium was removed, cells were incubated in MEM containing 5% FCS for 24 h. The medium was then removed, extracted with ethyl acetate and analyzed by reverse-phase HPLC. In (A) the presence of a major peak corresponding to that previously reported for 16:3 [5], more polar than arachidonic acid. In (B) only the peak corresponding to AA was detected.

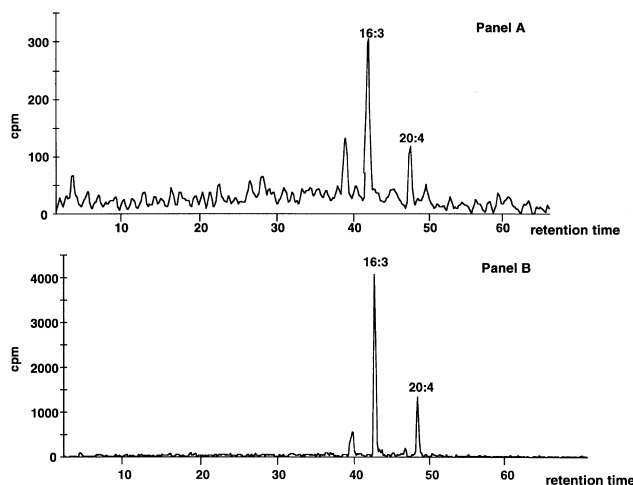


Fig. 3. (A,B) HPLC analysis of arachidonic acid metabolites released by fibroblasts from Alzheimer disease (A) and X-ALD (B). Fibroblasts were incubated with [^3H]arachidonic acid under conditions identical to those described for Fig. 2.

apparatus connected with a radiodetector. The column used was an RP-Nucleosil C18, the flow was 1 ml/min, and the solvent system was acetonitrile/water (80:20 v/v) [13].

The conversion of 20:5 n-3 (EPA) to the longer chain, more unsaturated products, including 22:6 (DHA), by fibroblasts, was assessed by HPLC analysis of products released into the medium, after cell incubation with [$1-^{14}\text{C}$]EPA (0.25 $\mu\text{Ci/ml}$) for 18 h. In ZW the conversion of EPA to 22:5 was 50–60% (Fig. 1, panel B), while in controls the conversion was only 30% (panel A). In addition, in ZW, but not in all the other samples (controls, Alzheimer and X-ALD samples), there was a peak, accounting for about 3–5% of the radioactivity, with the chromatographic characteristics of 24:6 [12], i.e. the substrate of peroxisomal β -oxidation in the formation of 22:6. In all samples except ZW, a peak corresponding to 22:6 n-3, containing a little over 1% of the total radioactivity, was detected, whereas in ZW not even a trace of this compound was found.

The incubation of skin fibroblasts from all subjects except ZW with [^3H]AA, following the described procedure, resulted in the formation of a compound detected by radio HPLC, as a peak more polar than AA (Fig. 2A). The observed compound had the same chromatographic properties of 4,7,10-hexadecatrienoic acid (16:3) [5]. When the same cells were incubated with [$1-^{14}\text{C}$]AA the peak was not present (Fig. 2B), due to the loss of the labeled carboxyl group resulting from a β -oxidation process. The same incubations were carried out in fibroblasts obtained from patients affected by Alzheimer's diseases and X-ALD and in both cases the production of 16:3 (Fig. 3A,B) was observed.

When ZW fibroblasts were incubated with [^3H]AA, the above compound was not detected in the HPLC profile (Fig. 4). Interestingly HPLC analyses showed also the accumulation of two major metabolites, less polar than AA, one of them identified as docosatetraenoic acid (22:4), with the

retention time of 52.3 min (Fig. 4). The conversion of AA to 22:4 n-6 amounted to about 35–40% and 15% of the substrate was converted to the other peak, versus about 10–12% conversion to 22:4 n-6 and traces to the second peak in control fibroblasts (Fig. 2A). The total conversion of AA, evaluated by the residual radioactivity associated with AA at the end of the incubation, was substantially identical in control and ZW fibroblasts, amounting to about 60%, at 18 h of incubation.

The results obtained in experiments on the conversions of AA and EPA in fibroblasts from controls and ZW patients are summarized in Table 2. Values in ZW are distinctly different from all the others and rather uniform within the group, since no 16:3 was produced from AA, versus a markedly greater production of the elongation product 22:4. Both fatty acids were detected in the medium as free acids. The conversion of EPA to 22:5 was also markedly greater in ZW, with no detectable formation of 22:6, when compared to all the other samples. The fatty acids derived through desaturation and elongation of EPA, detected as methyl esters prepared from cell lipids, were present in cells in esterified form.

Our study confirms severe abnormalities in the oxidative metabolism of AA and in the formation of DHA in ZW fibroblasts, while detecting complex associated changes in the profiles of intermediates in the metabolism of these fatty acids, which are consistent with the basic peroxisomal defect in this disease.

In fact, in ZW incubated with [$1-^{14}\text{C}$]EPA, the accumulation of 22:5 n-3, its elongated product, and of 24:6 n-3, the product of subsequent elongation and $\Delta 6$ desaturation (not present in controls), together with the complete absence of 22:6 n-3, can be attributed to the blockade of the (peroxisomal) β -oxidation of 24:6 to 22:6.

The abnormalities in AA β -oxidation are responsible for the lack of production of the 16:3 metabolite, on one side [5], and, on the other, for the accumulation of the elongation/desaturation products of the substrate, also previously described and discussed [5]. In addition, the alterations were specific for ZW, not being observed in the other types of samples (X-ALD and Alzheimer's Disease). In ZW, blockade of peroxisomal β -oxidation appeared to redirect the conversion of AA through the elongation/desaturation path-

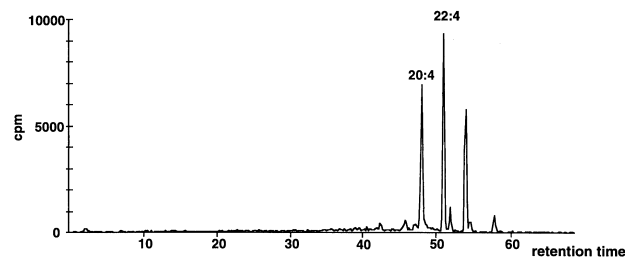


Fig. 4. A typical HPLC profile of arachidonic acid metabolites released by fibroblasts from a ZW patient. Fibroblasts were incubated with [^3H]arachidonic acid under conditions identical to those described for Fig. 2.

Table 2

Conversion of 20:4 n-6 and of 20:5 n-3 to various products in skin fibroblasts from controls (C), Zellweger (ZW), X-linked adrenoleukodystrophy (X-ALD) and Alzheimer disease (AD) patients

Samples	Substrates			
	20:4 n-6		20:5 n-3	
	Metabolites ^a		Metabolites ^b	
	16:3	22:4	22:5	22:6
C 1	49	12	33	1.4
C 2	47	12	32	1.4
C 3	48	12	31	1.0
ZW 1	0	39	53	0
ZW2	0	39	56	0
ZW3	0	32	67	0
ALD	53	11	32	1.2
AD1	50	12	32	1.4
AD2	51	12	33	1.3

Values represent the percentage of incorporation of radioactivity in the indicated compounds, after 18 h incubation, and each value is the average of duplicate determinations. ^aDeterminations directly made on lipids extracted from the medium of incubation. ^bDeterminations made on lipids extracted from cells and subjected to trans-methylation.

way with consequent accumulation of the 22:4 product. This could be further favoured by blockade of the β -oxidation also of this long-chain PUFA. Redirection of AA metabolism toward elongation/desaturation rather than β -oxidation in ZW fibroblasts is supported also by the observation that total AA conversion was substantially identical in ZW and in the other samples (about 60% of the substrate, at 18 h). As already proposed [5], AA does not have to undergo β -oxidation to form radioactive 22:4, since this product was detected both when the substrate was [³H] or [1-¹⁴C].

The formation of 16:3 from AA in X-ALD fibroblasts indicates that the β -oxidation of this 20 carbon PUFA is not impaired, in spite of the well known accumulation of the 24 and 26 carbon saturated fatty acids, attributed to reduced oxidation, in plasma and cell lipids of these patients. A defect in X-ALD concerns the peroxisomal VLC acyl-CoA ligase (24:0-CoA ligase), while oxidations of 22:1 n-9 and 22:4 n-6 (depressed in neonatal ALD [2]), and of AA and EPA are normal [2,15]. In Alzheimer's disease, as expected, there was no overt alteration of AA metabolism.

In conclusion, the pathway for the synthesis of DHA from EPA is impaired in steps beyond the formation of the intermediates with 22 and 24 carbon atoms. These alterations are specific for ZW. In addition, the reported results confirm that skin fibroblasts of three Italian ZW patients are unable to generate the product of AA β -oxidation 16:3 and that this is associated with a redirected pathway, resulting

in accumulation of the longer chain, more unsaturated derivative.

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